

An Overview of the Hyperledger Sawtooth Framework

Hyperledger Sawtooth is a modular platform for building, deploying and running distributed ledgers, which provide digital records (such as asset ownership) that are maintained without a central authority or implementation. Let's take a quick look at it.



Hyperledger Sawtooth is a blockchain framework that uses a modular platform for building, deploying and running distributed ledgers. Such distributed ledger solutions can use various consensus algorithms, depending on the size of the network. Initially contributed by Intel, Hyperledger Sawtooth uses the Proof of Elapsed Time (PoET) consensus algorithm, which provides the scalability of the Bitcoin blockchain without the high energy consumption. PoET allows for a highly scalable network of validator nodes. Hyperledger Sawtooth is designed for versatility, with support for both permissioned and permissionless deployments.

Sawtooth is the only Hyperledger project that provides the distributed state agreement, which means that you can trust every node in the system to understand information in the same way. It is the only framework that provides the ability to create adapters for any kind of transaction logic which means, for instance, that you can now run Ethereum Virtual Machine code, like Solidity, compile it and then run that on a Sawtooth based network.

The functional elements of Hyperledger Sawtooth

Transaction validators validate transactions. In Hyperledger Sawtooth, validators apply blocks that cause a change in the state. More specifically, validators validate transaction blocks and ensure that transactions result in state changes that are consistent across all participants in the network.

Transaction families are smart contracts in Hyperledger Sawtooth. They define the operations that can be applied to transactions. This allows for flexibility in the level of versatility and risk that exists on a network. Transaction families are often called 'safer' smart contracts, because they specify a predefined set of acceptable smart contract templates, as opposed to programming smart contracts from scratch. Hyperledger Sawtooth's transaction families can be written in many languages, including JavaScript, Java, C++, Python and Go, which allows flexibility for businesses to bring their own transaction families. Transaction families consist of both transaction processors (the server-side logic) and clients (for use from Web or mobile applications).